

Synthesis Matrix Template

Literature Review Topic: Serious Games for Enhancing Emotional Intelligence in Children.

	Source / Paper #1	Source / Paper #2	Source / Paper #3	Source / Paper #4	Source / Paper #5	Source / Paper #6	Source / Paper #7
Name and Year	Improving early childhood Emotional Intelligence through traditional 'Balago' games in Kindergartens.	Effect of a Serious Game Jestimule in Emotional Recognition of Moroccan Autistic Children.	Adoption of Serious Game in the Developing of Emotional Intelligence Skills.	Serious game the Emotion Detectives helps to improve social-emotional skills of children with neurodevelopmental disorders.	Serious Games for Emotional Intelligence's Skills Development for Inner Balance and Quality of Life.	Using Serious Games to Train Adaptive Emotional Regulation Strategies.	Children's Perspectives on Using Serious Games as a Complement to Promoting Their Social-Emotional Skills.
Research Questions	1. How does the implementation of traditional 'Balago' games affect the emotional intelligence of children of age 5-6 in kindergarten? 2. What are the specific aspects of emotional intelligence that improve through the use of 'Balago' games?	1. How does the serious game 'JeStiMule' influence the ability of autistic children to recognize facial emotions. 2. What specific emotions are most and least recognized by children with ASD after engaging with the game? 3. How do variables such as age, reading ability and type of autism affect the recognition of emotions?	1. What is the relationship between student's emotional intelligence (EI) skills and their performance in a serious game designed for developing these skills? 2. How do contextual variables such as course of study, gender, age and years of professional experience influence student's performance in emotional intelligence within the game? 3. How effective is the FLIGBY serious game in enhance emotional intelligence skills among higher education students?	1. Can the Emotion Detection game improve emotion discrimination skills in children with neurodevelopmental disorders? 2. How do children with neurodevelopmental disorders perform in emotion recognition tasks compared to typically developing peers? 3. What are the reported behavioural improvements in daily life after the intervention?	1. How do serious games contribute to the development of emotional intelligence skills? 2. What specific emotional skills can be enhanced through serious games? 3. How effective are serious games in improving emotional intelligence in individuals with special educational needs?	1. Can serious games effectively train emotional regulation (ER) strategies in adolescents. 2. How do adolescents perceive the usability and effectiveness of an ecological momentary assessment (EMA) tool compared to traditional paper questionnaires for monitoring emotions? 3. What is the potential of the GameTeen system as a therapeutic tool for early detection of maladaptive ER patterns and Prevention of emotional disorders?	1. How do children perceive the usefulness of serious games in promoting their social-emotional skills? 2. What differences exist in children's satisfaction between sessions that used serious games and those that did not? 3. How do serious games contribute to children's overall satisfaction with the 'Me and Us of Emotions' program?
Methodology	Participants: Children aged 5-6 years from Darma Bahagia Kindergarten in Samarinda. Data Collection: Emotional Intelligence was assessed through pretests and	Participants: The study involved 17 autistic children aged 5 to 14, with different levels of language and academic skills. The sample had	Participants: The study involved 49 students from multidisciplinary backgrounds, primarily in Management and Computer Engineering with a majority being	Participants: 30 children with neurodevelopmental disorders (autism, ADHD, developmental language disorder) and 20 control children, plus 106	Literature review: Analysis of 14 scientific articles published between 2010 and 2021 that focused on serious games and emotional intelligence.	GameTeen System: A virtual multiplatform system with 3 components; Mood Induction Games, ER Strategies Training and EMA Tool.	Participants: 232 Children aged 8-12 years (M=9.09, SD=0.80), with a balanced gender distribution (122 boys and 110 girls).

	posttests measuring various aspects such as self awareness, emotion management, motivation, empathy and social relationships. Intervention: The experimental group engaged in traditional Balogo games, while a control group did not receive any treatment. Analysis: The effectiveness of the intervention was evaluated by comparing pretest and posttest scores, with the N-Gain method used to measure improvement.	the boy's dominance with 88% of boys. Tools and Procedures: CARS was used for diagnosis. Participants engaged with 'JeStiMulE' for one hour daily over three months, supervised by educators. The game included mini games to train emotion recognition skills, progressing from basic emotions to more complex expressions. Data Analysis: Statistical analyses were conducted using descriptive statistics and Spearman Rank Order Correlations to evaluate the effectiveness of the game and the relationship between recognition skills and participant characteristics.	male and under 35 years old. Game Used: The FLIGBY serious game, based on Flow theory, was employed to facilitate the development of emotional intelligence skills. The game includes 21 scenarios requiring players to make decisions that impact employee motivation and performance, simulating a business environment. Data Collection: A mixed methods approach was utilized to assess the relationship between emotional intelligence and game performance, incorporating both quantitative metrics (game performance scores) and qualitative feedback from players. Performance Assessment: Players received immediate feedback from a virtual coach within the game, aiding in reflective learning and decision-making throughout the gameplay.	typically developing peers. Design: ABA research design with pre and post intervention assessments. Intervention: Children played the Emotion Detective game for two months, completing six emotion discrimination tasks.	Data Sources: Research conducted using databases like Medline, Science Direct and ACM Digital Library.	Usability Study: 21 adolescents (12-14 years old) used EMA or paper questionnaires twice daily for 2 weeks to report emotions.	Program: 'Me and Us of Emotion's program, along with 10 sessions with serious games integrated into 5 of them' Data Collection: Quantitative measures of satisfaction after each session and qualitative feedback through open-ended questions. Analysis: Statistical analyses using t-tests and regression, along with thematic analysis of qualitative data.
Key Findings	Improvement in Emotional Intelligence: The experimental group showed a significant increase in emotional intelligence, with an average N-Gain of 76%, categorized as a high level	Emotion Recognition Improvement: Children showed significant improvement in recognizing emotions. The most recognized emotions	Significant correlation between EI skills and game performance. Contextual variables did not significantly alter the relationship.	Emotional Children in the intervention group showed significant improvement in emotion discrimination skills in four tasks while the	Serious games are effective in cultivating emotional intelligence skills, which results in quality of life. The review identified various games targeting	EMA vs Paper Questionnaires: Adolescents found EMA more comfortable and useful than paper questionnaires. EMA's alarms helped them	No significant differences in satisfaction between sessions that used serious games and those that did not. Satisfaction with sessions using serious

	of improvement. This indicates that the Balogo games effectively enhanced the emotional intelligence of the children. Control Group Results: In contrast, the control group exhibited only a 16% N-Gain, categorized as low, demonstrating that without the intervention, emotional intelligence did not improve. Specific Aspects of Improvement: The study found notable advancements in various emotional intelligence aspects, particularly in self-awareness and motivation, suggesting that traditional games like Balogo can play a crucial role in holistic child development.	where happiness (69.34%), surprise (66.04%), and anger (10.30%) were the least recognized. Challenges Identified: Despite improvements, children struggled with recognizing neutral expressions, funny faces and disgust, indicating areas needing further intervention. Influence of Demographics: Correlation analysis revealed that age and reading ability negatively impacted emotion recognition, while the type of autism had a strong positive correlation with recognition skills. Effectiveness of 'JeStiMulE': The game was found to be an effective educational tool for enhancing emotion recognition in children with autism, suggesting potential for broader application in similar educational contexts. Further research is recommended to explore its effects on a larger scale and in different cultural settings.	FILIGBY effectively enhanced EI skills in the students and demonstrated its potential in the field of education.	control group improved in only one task. Improvements were maintained at one month follow up. Parents reported enhanced emotion recognition and behaviour skills in daily life for children in the intervention group compared to controls.	general emotional intelligence empathy and those tailored for individuals with special educational needs.	remember to report emotions. GameTeen Potential: The System shows promise as a therapeutic tool for training ER strategies and detecting maladaptive patterns early to prevent emotional disorders.	games was positively associated with fun, ease of understanding, and coping with emotions. Qualitative feedback revealed themes of positive aspects, negative aspects and opportunities for improvement indicating a general appreciation for serious games as a complementary tool in socio-emotional learning.
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